CS-GY 6923: Lecture 3
Regularization + Bayesian Perspective

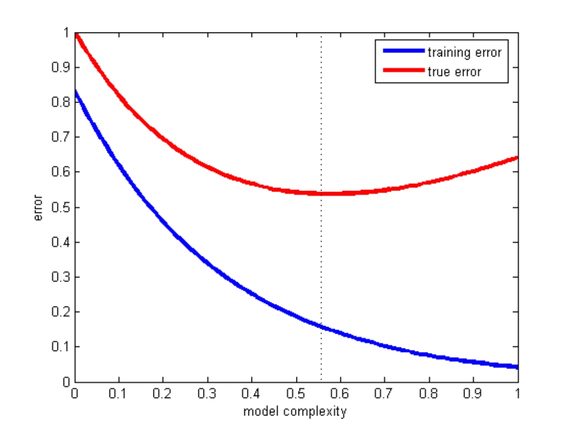
NYU Tandon School of Engineering, Akbar Rafiey

Course news

- The first written HW will be posted this weekend.
- 10% extra credit will be given if solutions are typewritten using LaTeX, Markdown, etc.
- All solutions must be written independently.
- Reminder: Lab 02 due September 24.

Recap on selecting model complexity

Model selection: How complex should our model be ?



Model selection 1: train-test paradigm

More reasonable approach: Evaluate model on fresh test data which was not used during training.

Test/train split:

- Given data set $(\mathbf{X}, \mathbf{y})$, split into two sets $(\mathbf{X}_{\text{train}}, \mathbf{y}_{\text{train}})$ and $(\mathbf{X}_{\text{test}}, \mathbf{y}_{\text{test}})$.
- Train q models $f^{(1)}, \ldots, f^{(q)}$ of varying complexity by finding parameters which minimize the loss on $(\mathbf{X}_{\text{train}}, \mathbf{y}_{\text{train}})$.
- Evaluate loss of each trained model on $(\mathbf{X}_{\text{test}}, \mathbf{y}_{\text{test}})$.
- Pick model with lowest test loss.

Sometimes you will see the term **validation set** instead of test set. Sometimes there will be both: use validation set for choosing the model, and test set for getting a final performance measure.

Train-test intuition

Is "test error" the end goal though? Don't we care about "future" error?

Intuition: Models which perform better on the test set will **generalize** better to future data.

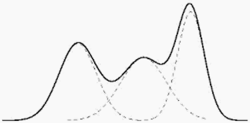
Goal: Introduce a little bit of formalism to better understand what this means. What is "future" data?

Statistical learning model

Statistical Learning Model:

- Assume each data example is randomly drawn from some distribution $(\mathbf{x}, y) \sim \mathcal{D}$.

E.g. $x_1, \ldots, x_d$ are Gaussian random variables with parameters $\mu_1, \sigma_1, \ldots, \mu_d, \sigma_d$.



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This is not really a simplifying assumption! The distribution could be arbitrarily complicated.

Risk

Statistical Learning Model:

- Assume each data example is randomly drawn from some distribution $(\mathbf{x}, y) \sim \mathcal{D}$.
- Define the **Risk** of a model/parameters:

$$R(f, \boldsymbol{\theta}) = \mathbb{E}_{(\mathbf{x},y)\sim\mathcal{D}}\left[L\left(f(\mathbf{x}, \boldsymbol{\theta}), y\right)\right]$$

  here L is our loss function (e.g. $L(z, y) = |z - y|$ or $L(z, y) = (z - y)^2$).

Ultimate Goal: Find model $f \in \{f^{(1)}, \ldots, f^{(q)}\}$ and parameter vector $\boldsymbol{\theta}$ to minimize the $R(f, \boldsymbol{\theta})$.

- **(Population) Risk**:

$$R(f, \boldsymbol{\theta}) = \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}} \left[L\left(f(\mathbf{x}, \boldsymbol{\theta}), y\right) \right]$$

- **Empirical Risk**: Draw $(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n) \sim \mathcal{D}$

$$R_E(f, \boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^{n} L\left(f(\mathbf{x}_i, \boldsymbol{\theta}), y_i\right)$$

Empirical risk

$$\mathbb{E}\left[R_E(f,\theta)\right] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} L(f(x_i;\theta), y_i)\right]$$

$$\downarrow$$

$$\varnothing$$

$$= \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}\left[L(f(x_i,\theta), y_i)\right] = \mathbb{E}_{(x,y)\sim D}\left[L(f(x,\theta), y)\right]$$

$$= R(f,\theta)$$

For any fixed model f and parameters $\boldsymbol{\theta}$,

$$\mathbb{E}\left[R_E(f,\boldsymbol{\theta})\right] = R(f,\boldsymbol{\theta}).$$

Only true if f and $\boldsymbol{\theta}$ are chosen *without looking at the data used to compute the empirical risk.*

Model selection

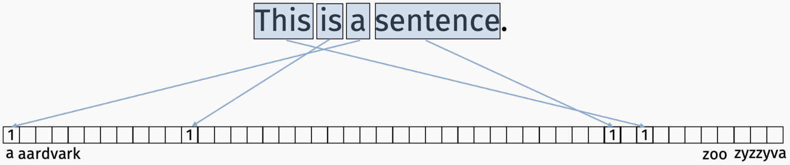
- Train q models $(f^{(1)}, \boldsymbol{\theta}_1^*), \ldots, (f^{(q)}, \boldsymbol{\theta}_q^*)$.
- For each model, compute empirical risk $R_E(f^{(i)}, \boldsymbol{\theta}_i^*)$ using test data.
- Since we assume our original dataset was drawn independently from $\mathcal{D}$, so is the random test subset.

No matter how our models were trained or how complex they are, $R_E(f^{(i)}, \boldsymbol{\theta}_i^*)$ is an unbiased estimate of the true risk $R(f^{(i)}, \boldsymbol{\theta}_i^*)$ for every i. Can use it to distinguish between models.

Model selection 2: bag of words

bag-of-words models and **n-grams**

Common way to represent documents (emails, webpages, books) as numerical data. The ultimate example of 1-hot encoding.



bag-of-words

Model selection example

bag-of-words models and **n-grams**

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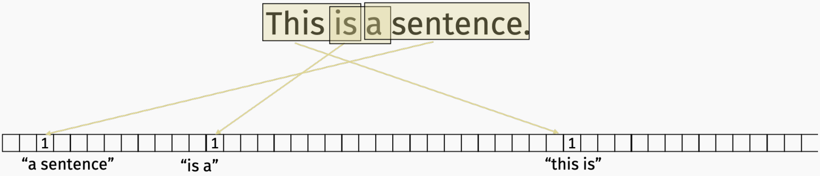
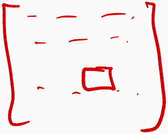
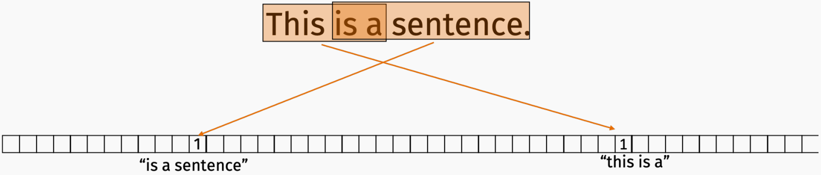


bi-grams



bag-of-words models and **n-grams**

Common way to represent documents (emails, webpages, books) as numerical data. The ultimate example of 1-hot encoding.



tri-grams

Model selection example

Models of increasing order:

- Model $f^{(1)}_{\boldsymbol{\theta}_1}$: spam filter that looks at **single words**.
- Model $f^{(2)}_{\boldsymbol{\theta}_2}$: spam filter that looks at **bi-grams**.
- Model $f^{(3)}_{\boldsymbol{\theta}_3}$: spam filter that looks at **tri-grams**.
- ...

"interest" "low interest" "low interest loan"

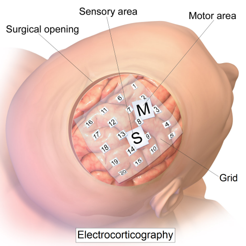
Increased length of **n-gram** means more expressive power.

Will be very relevant in our lab on generative language models!

Model selection 3: autoregressive models

Electrocorticography ECoG (Lab 02):

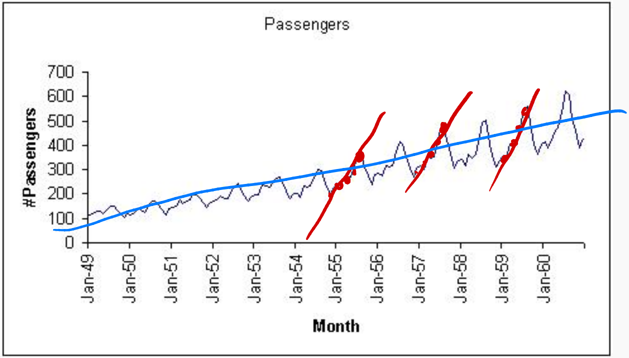
- Implant grid of electrodes on surface of the brain to measure electrical activity in different regions.



- Predict hand motion based on ECoG measurements.
- **Model order:** predict movement at time t using brain signals at time $t, t-1, \ldots, t-q$ for varying values of q.

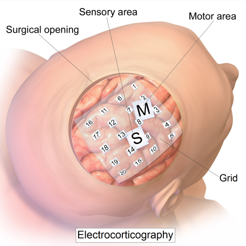
Autoregressive model

Predicting time t based on a linear function of the signals at time $t, t-1, \ldots, t-q$ is not the same as fitting a line to the time series. It's much more expressive.



Model selection lab tip

Electrocorticography ECoG lab:



First lab where computation actually matters (solving regression problems with $\sim 40k$ examples, ~ 1500 features)

Makes sense to test and debug code using a subset of the data.

Adaptive data analysis

Slight caveat: **The train-test paradigm is typically not how machine learning or scientific discovery works in practice!**

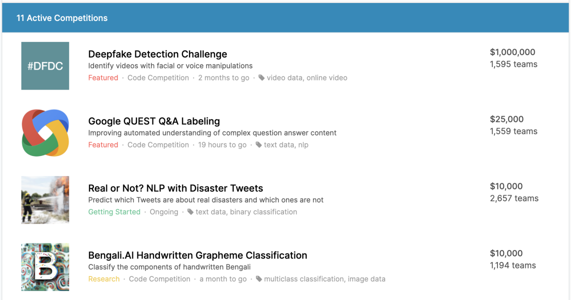
Typical workflow:

- Train a class of models.
- Test.
- Adjust class of models.
- Test.
- Adjust class of models.
- Cont...

Final model implicitly depends on test set because performance on the test set guided how we changed our model.

Adaptive data analysis

Popularity of ML benchmarks and competitions leads to adaptivity at a massive scale.



Kaggle (various competitions)

Imagenet (image classification and categorization)

Is adaptivity a problem? Does it lead to over-fitting? How much? How can we prevent it?

REPORT

The reusable holdout: Preserving validity in adaptive data analysis

Cynthia Dwork[1,*], Vitaly Feldman[2,*], Moritz Hardt[3,*], Toniann Pitassi[4,*], Omer Reingold[5,*], Aaron Roth[6,*]

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Do ImageNet Classifiers Generalize to ImageNet?

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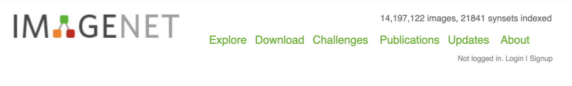
Rebecca Roelofs — UC Berkeley
Ludwig Schmidt — UC Berkeley
Vaishaal Shankar — UC Berkeley

12 Jun 2019

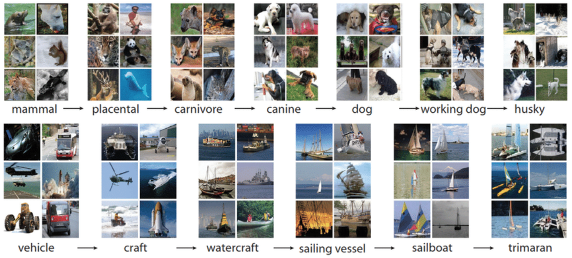
Abstract

We build new test sets for the CIFAR-10 and ImageNet datasets. Both benchmarks have been the focus of intense research for almost a decade, raising the danger of overfitting to excessively re-used test sets. By closely following the original dataset creation processes, we test to what extent current classification models generalize to new data. We evaluate a broad range of models and find accuracy drops of 3% – 15% on CIFAR-10 and 11% – 14% on ImageNet. However, accuracy gains on the original test sets translate to larger gains on the new test sets. Our results suggest that the accuracy drops are not caused by adaptivity, but by the models' inability to generalize to slightly "harder" images than those found in the original test sets.

Imagenet dataset

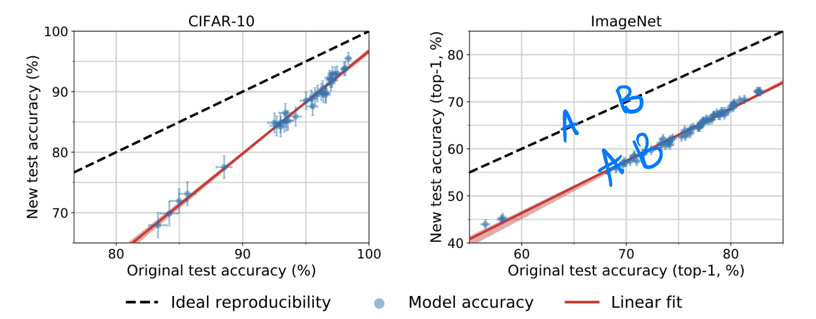


Collected by Fei-Fei Li's group at Stanford in 2006ish and labeled using Amazon Mechanical Turk.



We now have neural network models that can solve these classification problems with $>95\%$ accuracy.

Do ImageNet Classifiers Generalized to ImageNet?



Interestingly, when comparing popular vision models on "fresh" data, while performance dropped across the board, the relative rank of model performance did not change significantly.

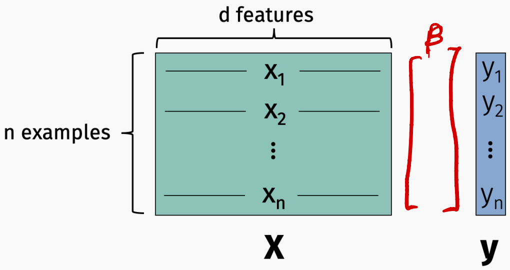
Regularization:
another way of controlling model complexity

Over-parameterized models

- In the model selection examples we discussed so far, we had full control over the complexity of the model: could range from underfitting to overfitting.
- In practice, you often don't have this freedom. Even the most basic model might lead to overfitting.

Over-parameterized models

Example: Linear regression model where $d \geq n$.



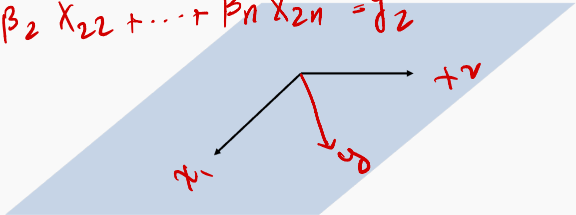
Can (almost) always find β so that $\mathbf{X}\beta = \mathbf{y}$ exactly.

Why?

High dimensional linear models

Claim: For almost all sets of n, length n vectors $\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)} \in \mathbb{R}^n$, we can write any vector $\mathbf{y}$ as a linear combination of these vectors.

$$\begin{cases} \beta_1 x_{11} + \beta_2 x_{12} + \cdots + \beta_n x_{1n} = y_1 \\ \beta_1 x_{21} + \beta_2 x_{22} + \cdots + \beta_n x_{2n} = y_2 \\ \vdots \end{cases}$$



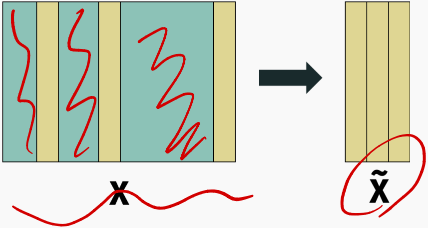
I.e., can find some coefficients so that
$\beta_1 \mathbf{x}^{(1)} + \ldots + \beta_{\cancel{d}\,n} \mathbf{x}^{(\cancel{d}\,n)} = \mathbf{X}\beta = \mathbf{y}$.

Zero train loss

- We will discuss some models later in the class where zero training loss is not necessarily a bad sign: k-nearest neighbors (the inductive bias of k-NN is local similarity), some neural nets (e.g., some empirical studies on double descent).
- Typically however it will be a sign of overfitting, as in the polynomial regression example.

Feature selection

Select some subset of $\ll n$ features to use in model:



Filter method: Compute some metric for each feature, and select features with highest score.

- Example: compute loss or R^2 value when each feature in $\mathbf{X}$ is used in single variate regression.

Any potential limitations of this approach?

Feature selection

Exhaustive approach: Pick best subset of q features. Train $\binom{d}{q}$ models.

$$O(d^d)$$

$$= \frac{d!}{q!\,(d-q!)}$$

$$= \frac{d(d-1)(d-2)\cdots\cancel{(d-q)!}}{q!\,\cancel{(d-q)!}}$$

Feature selection

Faster approach: Greedily select q features.

Stepwise Regression:

- **Forward:** Step 1: pick single feature that gives lowest loss. Step k: pick feature that when combined with previous $k-1$ chosen features gives lowest loss.
- **Backward:** Start with all of the features. Greedily eliminate those which have least impact on model performance.

Feature selection deserves more than two slides, but we won't go into too much more detail!

Alternative approach

Regularization: Discourage overfitting by adding a regularization penalty to the loss minimization problem.

$$\min_{\boldsymbol{\beta}} \left[L(\boldsymbol{\beta}) + Reg(\boldsymbol{\beta}) \right].$$

Alternative approach

Regularization: Discourage overfitting by adding a regularization penalty to the loss minimization problem.

β

$$\min_{\beta} \left[L(\beta) + Reg(\beta) \right].$$

Example: Least squares regression: $L(\beta) = \|\mathbf{X}\beta - \mathbf{y}\|_2^2$.

- Ridge regression (ℓ_2): $Reg(\beta) = \lambda \|\beta\|_2^2$ $\lambda \geq 0$
- LASSO (least absolute shrinkage and selection operator) (ℓ_1): $Reg(\beta) = \lambda \|\beta\|_1$ $\lambda \geq 0$
- Elastic net: $Reg(\beta) = \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|_2^2$

  $\lambda_1, \lambda_2 \geq 0$

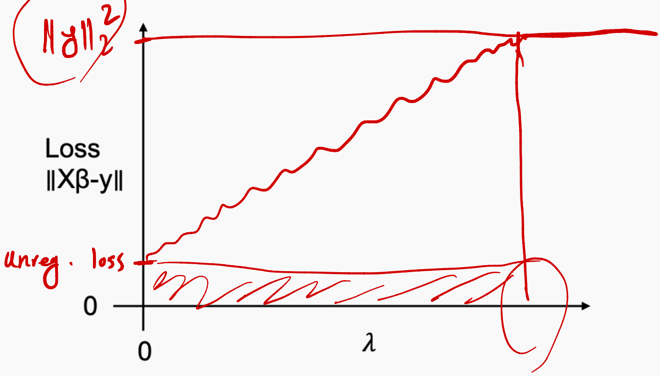
Note that $\arg\min_{\beta} [L(\beta) + Reg(\beta)] \neq \arg\min_{\beta} [L(\beta)]$

Ridge regularization: perspective 1

$= \lambda = \infty$

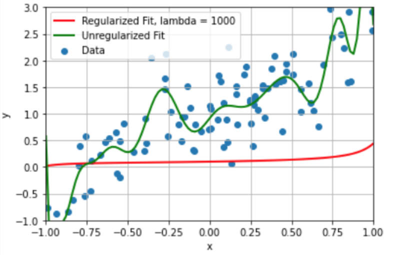
Ridge regression: $\min_{\beta} \left(\|\mathbf{X}\beta - \mathbf{y}\|_2^2 + \lambda\|\beta\|_2^2 \right).$

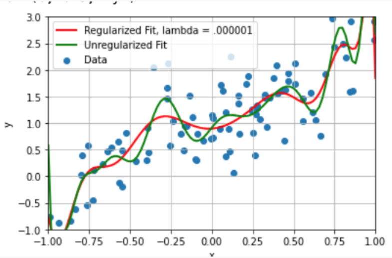
- As $\lambda \to \infty$, we expect $\|\beta\|_2^2 \to 0$ and $\|\mathbf{X}\beta - \mathbf{y}\|_2^2 \to \|\mathbf{y}\|_2^2$.
- By choosing different values of λ we have models of varying accuracy/complexity.

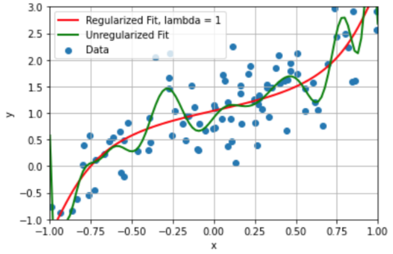


Polynomial examples

Fit degree 20 polynomial with varying levels of regularization.







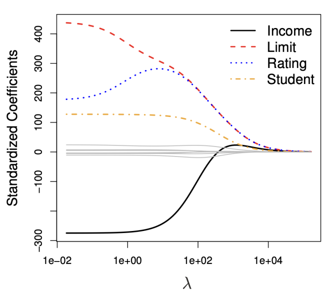
Ridge regularization: perspective 2

X β

- Ridge regression: $\min_\beta \left(\|\mathbf{X}\beta - \mathbf{y}\|_2^2 + \lambda\|\beta\|_2^2\right)$.

$\|\beta\|_2^2 = \beta_1^2 + \beta_2^2 + \cdots + \beta_d^2$

- As $\lambda \to \infty$, we expect $\|\beta\|_2^2 \to 0$ and $\|\mathbf{X}\beta - \mathbf{y}\|_2^2 \to \|\mathbf{y}\|_2^2$.
- Feature selection methods attempt to set many coordinates in β to 0. Ridge regularizations encourages coordinates to be close to zero.



Ridge regularization

How do we minimize: $L_R(\beta) = \|\mathbf{X}\beta - \mathbf{y}\|_2^2 + \lambda\|\beta\|_2^2$?

$$\nabla L_R(\beta) = 2X^T(X\beta - y) + 2\lambda\beta$$

$$= 2X^T X\beta - 2X^T y + 2\lambda\beta$$

$$\nabla L_R(\beta) = 0 \Rightarrow \cancel{2}X^T X\beta - \cancel{2}X^T y + \cancel{2}\lambda\beta = 0$$

$$\Rightarrow X^T X\beta + \lambda\beta = X^T y$$

$$\Rightarrow (X^T X + \lambda \underbrace{I}_{d \times d})\beta = X^T y$$

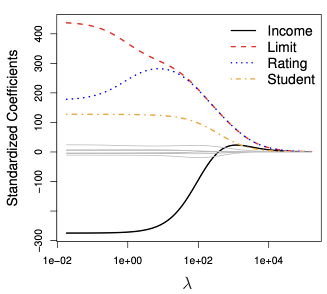
$$\Rightarrow \beta = (X^T X + \lambda I)^{-1} X^T y$$

Lasso regularization

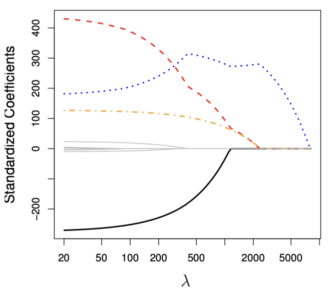
$$\text{Lasso regularization: } \min_\beta \|\mathbf{X}\beta - \mathbf{y}\|_2^2 + \lambda\|\beta\|_1.$$

$= |\beta_1| + |\beta_2| + \cdots + |\beta_d|$

- As $\lambda \to \infty$, we expect $\|\beta\|_1 \to 0$ and $\|\mathbf{X}\beta - \mathbf{y}\|_2^2 \to \|\mathbf{y}\|_2^2$.
- Typically encourages subset of β_i's to go to zero, in contrast to ridge regularization.

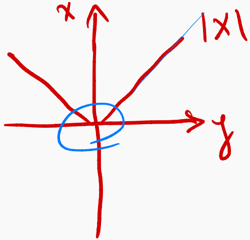


Ridge regularization



Lasso Regularization

Lasso regularization



Pros:

- Simpler, more interpretable model.
- More intuitive reduction in model order.

Cons:

- No closed form solution because $\|\beta\|_1$ is not differentiable.
- Can be solved with iterative methods, but generally not as quickly as ridge regression.

?

Regularization

$\lambda = 0.1$

$\lambda = 0.2$

$$\boxed{} \underbrace{+ \lambda \beta_1^2 + \lambda \beta_2^2 + \cdots + \lambda \beta_d^2}$$

Notes:

- Model selection/cross validation used to choose optimal scaling λ on $\lambda\|\boldsymbol{\beta}\|_2^2$ or $\lambda\|\boldsymbol{\beta}\|_1$.
- Often grid search for best parameters is performed in "log space". E.g. consider $[\lambda_1, \ldots, \lambda_q] = 1.5^{[-4,-3,-2,-1,-0,1,2,3,4]}$.
- Regularization methods are not invariant to data scaling. Typically when using regularization we mean center and scale columns to have unit variance.

The bayesian/probabilistic modeling perspective

Classification setup

- **Data Examples:** $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$
- **Target:** $y_1, \dots, y_n \in \{0, 1, \dots, q-1\}$ when there are q classes.
  - Binary Classification: $q = 2$, so each $y_i \in \{0, 1\}$.
  - Multi-class Classification: $q > 2$. [1]

[1] Note that there is also multi-label classification where each data example may belong to more than one class.

Classification examples

- Medical diagnosis from MRI: 2 classes.
- MNIST digits: 10 classes.
- Full Optical Character Regonition: 100s of classes.
- ImageNet challenge: 21,000 classes.

Running example today: Email Spam Classification.

Classification

Classification can (and often is) solved using the same **loss-minimization framework** we saw for regression.

We won't see that today! We're going to use classification as a window into another way of thinking about machine learning.

Will give an interesting new justifications for tools like regularization. Will also give us an approach for generative ML.

Rest of Today: ML from a **Probabilistic Modeling/Bayesian Perspective**.

Probabilistic modeling

In a Bayesian or Probabilistic approach to machine learning we always start by conjecturing a

probabilistic model

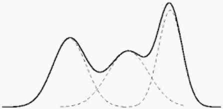
that plausibly could have generated our data.

- The model guides how we make predictions.
- The model typically has unknown parameters $\vec{\theta}$ and we try to find the most reasonable parameters based on **observed** data (more on this later in lecture).

Goal:

- Build a probabilistic model for a binary classification problem.
- Estimate parameters of the model.
- From the model derive a classification rule for future predictions (the **Naive Bayes Classifier**).

Typically we try to keep things simple!



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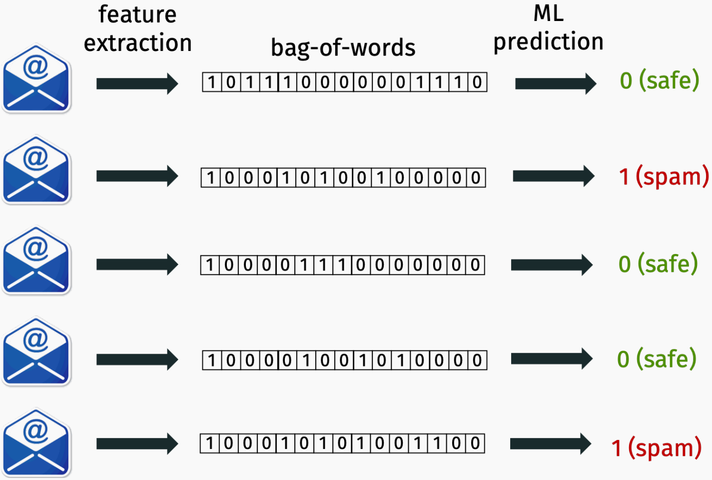
X

X

Spam prediction



Both target labels and data vectors are binary.

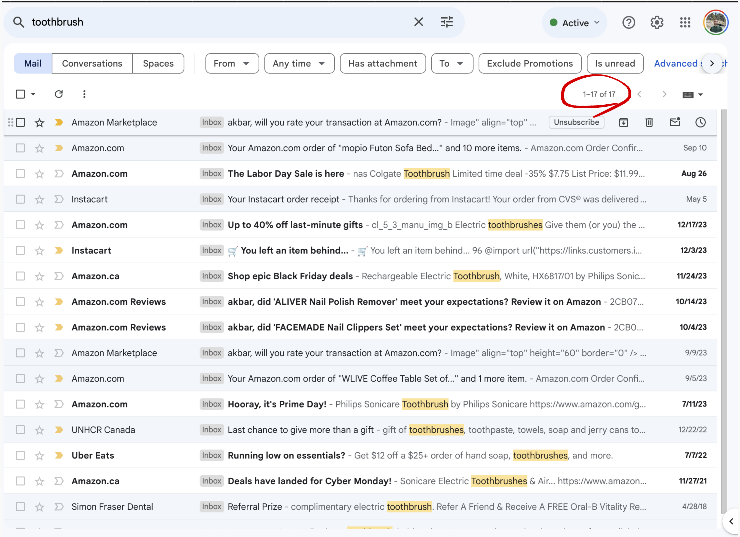
Email model

Let's create a model that generates spam and non-spam emails.
Observation: Since bag-of-words features don't care about word order, our model does not need to either.

- Common approach: assign a probability $p_i \in [0, 1]$ to word i. Set $\mathbf{x}_i = 1$ with probability p_i, $\mathbf{x}_i = 0$ with probability $1 - p_i$.

$$p_{\text{the}} = 0.99 \qquad p_{\text{calendar}} = 0.7 \qquad p_{\text{toothbrush}} = \frac{17}{11K}$$

Email model


toothbrush
Active
Mail
Conversations
Spaces
From
Any time
Has attachment
To
Exclude Promotions
Is unread
Advanced
1–17 of 17
Amazon Marketplace Inbox akbar, will you rate your transaction at Amazon.com? - Image" align="top" ... Unsubscribe
Amazon.com Inbox Your Amazon.com order of "mopio Futon Sofa Bed..." and 10 more items. - Amazon.com Order Confir... Sep 10
Amazon.com Inbox The Labor Day Sale is here - nas Colgate Toothbrush Limited time deal -35% $7.75 List Price: $11.99... Aug 26
Instacart Inbox Your Instacart order receipt - Thanks for ordering from Instacart! Your order from CVS® was delivered ... May 5
Amazon.com Inbox Up to 40% off last-minute gifts - cl_5_3_manu_img_b Electric toothbrushes Give them (or you) the ... 12/17/23
Instacart Inbox You left an item behind... - You left an item behind... 96 @import url("https://links.customers.i... 12/3/23
Amazon.ca Inbox Shop epic Black Friday deals - Rechargeable Electric Toothbrush, White, HX6817/01 by Philips Sonic... 11/24/23
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Email model

How can we make this model richer when we take spam into account?

- Different words tend to be more or less frequent in spam or regular emails.

Not Spam

$$p_{\text{won}} = 0.02$$
$$p_{\$} = 0.001$$
$$p_{student} = 0.7$$

Spam

$$p_{\text{won}} =$$
$$p_{\$} = 0.2$$
$$p_{student} = 0.4$$

Probabilistic model for email

Probabilistic model for (bag-of-words, label) pair $(\mathbf{x}, y)$:

- Set $y = 0$ with probability p_0, $y = 1$ with probability $p_1 = 1 - p_0$.
    - p_0 is probability an email is not spam (e.g. 99%).
    - p_1 is probability an email is spam (e.g. 1%).
- If $y = 0$, for each i, set $x_i = 1$ with prob. p_{i0}.
- If $y = 1$, for each i, set $x_i = 1$ with prob. p_{i1}.

Unknown model parameters:

- p_0, p_1,
- $p_{10}, p_{20}, \ldots p_{d0}$, one for each of the d vocabulary words.
- $p_{11}, p_{21}, \ldots p_{d1}$, one for each of the d vocabulary words.

How would you estimate these parameters?

Parameter estimation

Reasonable way to set parameters:

- Set p_0 and p_1 to the empirical fraction of not spam/spam emails.
- For each word i, set p_{i0} to the empirical probability word i appears in a non-spam email.
- For each word i, set p_{i1} to the empirical probability word i appears in a spam email.

Estimating these parameters from previous data examples is the only "training" we will do.

Done with modeling

on to prediction

Probability review

- **Probability:** p(x) – the probability event x happens.
- **Joint probability:** p(x,y) – the probability that event x and event y happen.
- **Conditional Probability** $p(x \mid y)$ – the probability x happens given that y happens.

$p(x,y) = p(x|y) \cdot p(y)$

$p(x,y) = p(y|x) \cdot p(x)$

$$p(x|y) = p(x,y) / p(y)$$

$$p(y|x) = p(x,y) / p(x)$$

$$p(x|y) = \frac{p(y|x)p(x)}{p(y)}$$

Proof:

$$p(x,y) = p(x|y) \cdot p(y)$$

$$p(x,y) = p(y|x) \cdot p(x)$$

$$\Rightarrow p(x|y) \cdot p(y) = p(y|x) \cdot p(x)$$

$$\Rightarrow p(x|y) = \frac{p(y|x) \cdot p(x)}{p(y)}$$

Classification rule

Given unlabeled input $(\mathbf{w}, \_\_\_)$, choose the label $y \in \{0, 1\}$ which is <u>most likely</u> given the data. Recall $\mathbf{w} = [0, 0, 1, \ldots, 1, 0]$.

Classification rule: maximum a posterior (MAP) estimate.

Step 1. Compute:

- $p(y = 0 \mid \mathbf{w})$: prob. $y = 0$ given observed data vector $\mathbf{w}$.
- $p(y = 1 \mid \mathbf{w})$: prob. $y = 1$ given observed data vector $\mathbf{w}$.

Step 2. Output: 0 or 1 depending on which probability is larger.

$p(y = 0 \mid \mathbf{w})$ and $p(y = 1 \mid \mathbf{w})$ are called **posterior** probabilities.

Evaluating the posterior

How to compute the posterior? **Bayes rule!**

$$p(y = 0 \mid \mathbf{w}) = \frac{p(\mathbf{w} \mid y = 0)p(y = 0)}{p(\mathbf{w})} \tag{1}$$

$$\text{posterior} = \frac{\text{likelihood} \times \text{prior}}{\text{evidence}} \tag{2}$$

- **Prior:** Probability in class 0 <u>prior</u> to seeing any data.
- **Posterior:** Probability in class 0 <u>after</u> seeing the data.

Evaluating the posterior

Goal is to determine which is larger:

$$p(y = 0 \mid \mathbf{w}) = \frac{p(\mathbf{w} \mid y = 0)p(y = 0)}{\cancel{p(\mathbf{w})}} \qquad \text{vs.}$$

$$p(y = 1 \mid \mathbf{w}) = \frac{p(\mathbf{w} \mid y = 1)p(y = 1)}{\cancel{p(\mathbf{w})}}$$

- We can ignore the evidence $p(\mathbf{w})$ since it is the same for both sides!
- $p(y = 0)$ and $p(y = 1)$ already known (computed from training data). These are our computed parameters p_0, p_1.
- $p(\mathbf{w} \mid y = 0) = ?$ $p(\mathbf{w} \mid y = 1) = ?$

Evaluating the posterior

the \$ student ...
Voc : [0 1 0]

$p(x,y) = p(x) \cdot p(y)$
if x, y are indep.

$w_0 \; w_1 \; w_2 \; w_3$

Consider the example $\mathbf{w} = [0, 1, 1, 0, 0, 0, 1, 0]$.

Recall that, under our model, index i is 1 with probability p_{i0} if we are not spam, and 1 with probability p_{i1} if we are spam .

$$p(\mathbf{w} \mid y = 0) =$$

$= P(w_0 \mid y=0) \cdot P(w_1 \mid y=0) \cdots p(w_7 \mid y=0)$

$= p(w_0 = 0 \mid y=0) \cdot p(w_1 = 1 \mid y=0) \cdots p(w_7 = 0 \mid y=0)$

$$p(\mathbf{w} \mid y = 1) =$$

$P(w_0=0 \mid y=1) \cdot P(w_1=1 \mid y=1) \cdots$

$= (1 - P_{01}) \cdot P_{11} \cdot P_{21} \cdots$

$= (1 - P_{00}) \cdot P_{10} \cdot P_{20} \cdot (1 - P_{30}) \cdots$

Naive bayes

Final Naive Bayes Classifier

Training/Modeling: Use existing data to compute:

- $p_0 = p(y = 0), p_1 = p(y = 1)$
- For all i compute:
  - $p_{i0} = p(x_i = 1 \mid y = 0)$ and $(1 - p_{i0}) = p(x_i = 0 \mid y = 0)$
  - $p_{i1} = p(x_i = 1 \mid y = 1)$ and $(1 - p_{i1}) = p(x_i = 0 \mid y = 1)$

Prediction:

$\begin{cases} \text{if } w_i = 1 \to p_{i0} \\ \text{if } w_i = 0 \to 1 - p_{i0} \end{cases}$

- For new input $\mathbf{w}$:
  - Compute $p(\mathbf{w} \mid y = 0) = \prod_i \overbrace{p(w_i \mid y = 0)}$
  - Compute $p(\mathbf{w} \mid y = 1) = \prod_i p(w_i \mid y = 1)$
- Return

$$\arg\max \left[p\left(\mathbf{w} \mid y = 0\right) \cdot p\left(y = 0\right), p\left(\mathbf{w} \mid y = 1\right) \cdot p\left(y = 1\right)\right].$$

Other applications of
the bayesian perspective

Bayesian regression

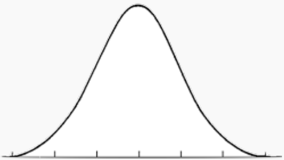
The Bayesian view offers an interesting alternative perspective on many machine learning techniques.

Example: Linear Regression.

Probabilistic model:

$$y = \langle \mathbf{x}, \boldsymbol{\beta} \rangle + \eta$$

where the η drawn from $N(0, \sigma^2)$ is **random Gaussian noise**.



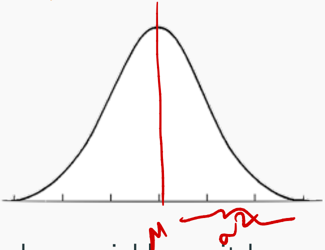
$$Pr(\eta = z) \sim e^{-z^2}$$

The symbol $\sim$ means "is proportional to".

Gaussian distribution refresher

Names for the same thing: Normal distribution, Gaussian distribution, bell curve.

Parameterized by mean μ and variance σ^2.



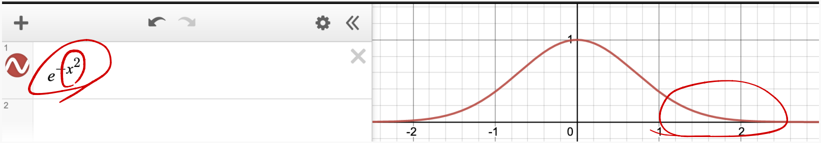
η is a continuous random variable, so it has a probability density function $p(\eta)$ with $\int_{-\infty}^{\infty} p(\eta) d\eta = 1$

$$p(\eta) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}(\frac{\eta-\mu}{\sigma})^2}$$

$e^{-\eta^2}$

Gaussian distribution refresher

The important thing to remember is that the the PDF falls off exponentially as we move further from the mean.



The normalizing constant in front 1/2, etc. don't matter so much.

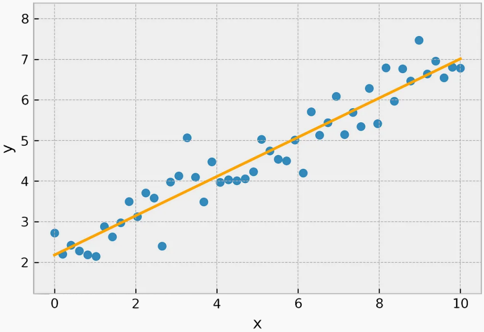
Illustration

Example: Linear Regression.

Probabilistic model:

$$y = \langle \mathbf{x}, \boldsymbol{\beta} \rangle + \eta$$

where the η drawn from $N(0, \sigma^2)$ is **random Gaussian noise**. The noise is independent for different inputs $\mathbf{x}_1, \ldots, \mathbf{x}_n$.



Bayesian regression

How should we select β for our model?

Also use a Bayesian approach!

Choose $\boldsymbol{\beta}$ to maximize:

$$\text{posterior} = \Pr(\boldsymbol{\beta} \mid \mathbf{X}, \mathbf{y}) = \frac{\Pr(\mathbf{X}, \mathbf{y} \mid \boldsymbol{\beta}) \Pr(\boldsymbol{\beta})}{\Pr(\mathbf{X}, \mathbf{y})} = \frac{\text{likelihood} \times \text{prior}}{\text{evidence}}.$$

In this case, we don't have a prior – no values of $\boldsymbol{\beta}$ are inherently more likely than others.

Choose $\boldsymbol{\beta}$ to maximize just the likelihood:

$$\frac{\Pr(\mathbf{X}, \mathbf{y} \mid \boldsymbol{\beta}) \cancel{\Pr(\boldsymbol{\beta})}}{\cancel{\Pr(\mathbf{X}, \mathbf{y})}} = \frac{\text{likelihood} \times \cancel{\text{prior}}}{\cancel{\text{evidence}}}.$$

This is called the **maximum likelihood estimate**.

Fixed design linear regression

Often we think of $\mathbf{X}$ as fixed and deterministic, and only $\mathbf{y}$ is generated at random in the model. This is called the fixed design setting. Can also consider a randomized design setting, but it is slightly more complicated.

In the fixed design setting our task of maximizing $\Pr(\mathbf{X}, \mathbf{y} \mid \boldsymbol{\beta})$ simplifies to maximizing

$$\max_{\boldsymbol{\beta}} \Pr(\mathbf{y} \mid \boldsymbol{\beta})$$

Maximum likelihood estimate

Data:

$$\mathbf{X} = \begin{bmatrix} — & \mathbf{x}_1 & — \\ — & \mathbf{x}_2 & — \\ & \vdots & \\ — & \mathbf{x}_n & — \end{bmatrix} \qquad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

Model: $y_i = \langle \mathbf{x}_i, \boldsymbol{\beta} \rangle + \eta_i$ where $p(\eta_i = z) \sim e^{-z^2/2\sigma^2}$ and $\eta_1, \ldots, \eta_n$ are independent.

$$\Pr(\mathbf{y} \mid \boldsymbol{\beta}) \sim$$

Log likelihood

Easier to work with the **log likelihood**:

$$\begin{aligned}
\arg\max_{\boldsymbol{\beta}} \Pr(\mathbf{X}, \mathbf{y} \mid \boldsymbol{\beta}) &= \arg\max_{\boldsymbol{\beta}} \prod_{i=1}^{n} e^{-(y_i - \langle \mathbf{x}_i, \boldsymbol{\beta}\rangle)^2/2\sigma^2} \\
&= \arg\max_{\boldsymbol{\beta}} \ \log\left(\prod_{i=1}^{n} e^{-(y_i - \langle \mathbf{x}_i, \boldsymbol{\beta}\rangle)^2/2\sigma^2}\right) \\
&= \arg\max_{\boldsymbol{\beta}} \sum_{i=1}^{n} -(y_i - \langle \mathbf{x}_i, \boldsymbol{\beta}\rangle)^2/2\sigma^2 \\
&= \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} (y_i - \langle \mathbf{x}_i, \boldsymbol{\beta}\rangle)^2.
\end{aligned}$$

Maximum likelihood estimator

Conclusion: Choose $\boldsymbol{\beta}$ to minimize:

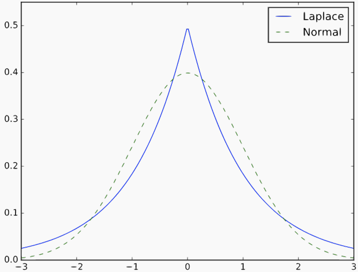
$$\sum_{i=1}^{n}(y_i - \langle \mathbf{x}_i, \boldsymbol{\beta} \rangle)^2 = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2.$$

This is a completely different justification for squared loss!

Minimizing the ℓ_2 loss is optimal in a certain sense when you assume your data follows a linear model with i.i.d. Gaussian noise.

Bayesian regression

If we had modeled our noise η as Laplace noise, we would have found that minimizing $\|\mathbf{y} - \mathbf{X}\beta\|_1$ was optimal.



$$Pr(\eta = z) \sim$$

Laplace noise has "heavier tails", meaning that it results in more outliers.

This is a completely different justification for ℓ_1 loss.

Bayesian regularization

We can add another layer of probabilistic modeling by also assuming $\boldsymbol{\beta}$ is random and comes from some distribution, which encodes our prior belief on what the parameters are.

Maximum a posteriori (MAP estimation):

$$\Pr(\boldsymbol{\beta} \mid \mathbf{X}, \mathbf{y}) = \frac{\Pr(\mathbf{X}, \mathbf{y} \mid \boldsymbol{\beta}) \Pr(\boldsymbol{\beta})}{\Pr(\mathbf{X}, \mathbf{y})}.$$

Assume values in $\boldsymbol{\beta} = [\beta_1, \ldots, \beta_d]$ come from some distribution.

- **Common model:** Each β_i drawn from $N(0, \gamma^2)$, i.e. normally distributed, independent.
- Encodes a belief that we are unlikely to see models with very large coefficients.

Bayesian regularization

Goal: choose $\boldsymbol{\beta}$ to maximize:

$$\Pr(\boldsymbol{\beta} \mid \mathbf{X}, \mathbf{y}) = \frac{\Pr(\mathbf{X}, \mathbf{y} \mid \boldsymbol{\beta}) \Pr(\boldsymbol{\beta})}{\Pr(\mathbf{X}, \mathbf{y})}.$$

- We can still ignore the "evidence" term $\Pr(\mathbf{X}, \mathbf{y})$ since it is a constant that does not depend on $\boldsymbol{\beta}$.
- $\Pr(\boldsymbol{\beta}) = \Pr(\beta_1) \cdot \Pr(\beta_2) \cdot \ldots \cdot \Pr(\beta_d)$
- $\Pr(\boldsymbol{\beta}) \sim$

Bayesian regularization

Easier to work with the **log**:

$$\begin{aligned}
&\arg\max_{\boldsymbol{\beta}} \Pr(\mathbf{X}, \mathbf{y} \mid \boldsymbol{\beta}) \cdot \Pr(\boldsymbol{\beta}) \\
&= \arg\max_{\boldsymbol{\beta}} \prod_{i=1}^{n} e^{-(y_i - \langle \mathbf{x}_i, \boldsymbol{\beta}\rangle)^2/2\sigma^2} \cdot \prod_{i=1}^{n} e^{-(\beta_i)^2/2\gamma^2} \\
&= \arg\max_{\boldsymbol{\beta}} \sum_{i=1}^{n} -(y_i - \langle \mathbf{x}_i, \boldsymbol{\beta}\rangle)^2/2\sigma^2 + \sum_{i=1}^{d} -(\beta_i)^2/2\gamma^2 \\
&= \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} (y_i - \langle \mathbf{x}_i, \boldsymbol{\beta}\rangle)^2 + \frac{\sigma^2}{\gamma^2} \sum_{i=1}^{d} (\beta_i)^2/\sigma^2.
\end{aligned}$$

Choose $\boldsymbol{\beta}$ to minimize $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \frac{\sigma^2}{\gamma^2}\|\boldsymbol{\beta}\|_2^2$.

Completely different justification for ridge regularization!

Bayesian regularization

Test your intuition: What modeling assumption justifies LASSO regularization: $\min \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda\|\boldsymbol{\beta}\|_1$?